

s1.02 — Software load balancer — L4 / L7

Index

- [Interview framing](#)
- [Requirements — IC3 baseline](#)
- [Capacity and consequence — IC3 baseline](#)
- [Architecture and flow — IC3 baseline](#)
- [API and data model — IC3 baseline](#)
- [Selection and concurrency — IC3 baseline](#)
- [Health, retries, and draining — IC3 baseline](#)
- [Failure, operations, and security](#)
- [Alternatives and IC4 stretch](#)
- [Level signal — IC3 baseline vs IC4 stretch](#)
- [Interview pushback and validation](#)
- [Recall card](#)
- [References](#)

Interview framing

- Target: OCI IC3 ownership of a bounded platform component.
 - IC3 baseline is what must land in every phase.
 - IC4 stretch adds architecture changes under stronger constraints.
 - These are preparation labels, not Oracle's official rubric.
- Evidence: retain the first-hand report attribution in [the bank](#).
 - The architecture below is a proposed interview answer.
 - It does not claim to reproduce Oracle's implementation or an interviewer's accepted answer.
- Why this is hard
 - Naive: round robin over a static backend list behind one proxy process.
 - First break: one slow backend absorbs its round-robin share and stalls
 - A failed health probe on every backend removes the whole pool.
 - Second break: a timed-out POST is retried onto a second backend, a deploy drains a backend that still owns long streams, and a configuration push lands on half the fleet.
 - Minimum primitives
 - An immutable configuration snapshot per request
 - A local eligible set with in-flight counts
 - Health with hysteresis separate from administrative state
 - A retry budget.
- 🗣️ "Balancing is the easy part; the design is what the proxy does when its evidence about backends is stale, partial, or wrong."
- Opening script
 - "I will start with an L7 HTTP reverse proxy, trace one request, and keep routing configuration separate from connection state."
 - "First I confirm the protocol, whether TLS terminates at the proxy, whether writes are retry-safe, and whether session affinity is required."
 - "Then I size in-flight requests, buffers, and probe traffic, and define the control-plane API for pools, routes, and drains."
 - "The deep dive is backend selection under skew, health with hysteresis, the retry contract, and draining."
 - "I will close with failure behavior when health evidence is unreliable, and what changes at ten times the load."

Phase	Clock	Deliverable
Intro and behavioral slice	00:00–05:00	90-second pitch; confirm the prompt
Requirements	05:00–11:00	Protocol choice, clarifying answers, snapshot invariant
Estimates and contract	11:00–17:00	In-flight and buffer sizing; pool, route, and drain API
Architecture	17:00–30:00	Diagram, control/data plane split, one request trace
Deep dive	30:00–45:00	Power-of-two selection, health hysteresis, retry contract, drain
Operations and tradeoffs	45:00–50:00	Failure table, panic policy, tradeoff, wrap script
Wrap-up	50:00–54:00	Four-sentence recap and weakest assumption
Questions for interviewer	54:00–60:00	Two role-specific questions

- Clock note
 - If the interviewer states 45 minutes, drop the intro to 2 minutes and compress each phase proportionally.
 - The interviewer's clock wins over this table.

Requirements — IC3 baseline

- Numerical targets are interview assumptions unless an adjacent primary source states otherwise.

Question	Assumed answer	Why it changes the design
L4 or L7?	L7 HTTP reverse proxy ; L4 forwarding as the alternative	Decides whether routing sees host and path and whether TLS terminates at the proxy
Does TLS terminate at the proxy?	Yes, with certificate references from the PKI service	Proxies hold private keys and handshake CPU becomes a sizing input
Are writes safe to retry?	Only where the backend enforces idempotency	Bounds the retry contract to connect failures and idempotent requests
Is session affinity required?	No by default; opt-in per route	Affinity concentrates load and does not preserve backend memory
Fleet shape?	10 proxies, 500 backends, one region, multiple fault domains	Sets probe traffic and how much capacity one failure domain removes
Who distributes connections to proxies?	A redundant front door, managed or ECMP/VIP	Keeps the proxy fleet horizontally scalable without owning routing
Multi-tenant?	Yes; routes and pools are tenant-owned resources	Configuration is tenant-scoped and shared ejection has a blast radius

- Functional scope
 - **Route** by host and path.
 - **Balance** across eligible backends.
 - **Remove** unhealthy backends with hysteresis.
 - **Drain** connections for deployments.
 - **Apply validated configuration** without restarting the fleet.
 - Network layer 4 and layer 7 are unrelated to the depth labels in this file.

Requirement	Target	Enforced by
Peak throughput	100,000 requests/s	Fleet sizing on the measured limiting resource; front-door spread
Backend inventory	500 backends across pools	Bounded eligible-array snapshots per pool
Added proxy latency	p99 < 5 ms excluding backend execution	Constant-time selection; bounded buffering; no per-request coordination
Failure detection	Within 10 s under normal scheduling	Probe interval 2 s , timeout 0.5 s , three failures to eject
Regional request availability	99.95%	Redundant front door and proxy capacity sufficient after the largest planned fault-domain loss; cached configuration

Requirement	Target	Enforced by
Drain deadline	At most 30 s before terminating remaining connections	Drain operation with a deadline and forced close
Retry amplification	Retries at most 10% of admitted original requests	Per-proxy retry budget
Configuration convergence	Acknowledged fleet-wide within 60 s	Versioned push; <code>appliedVersion</code> visible per proxy

- Invariants
 - A request uses one internally consistent route/configuration snapshot.
 - No new request is deliberately assigned to a drained backend.
 - Retries cannot assume an unobserved response means the backend did no work.
 - Numerical targets above are interview assumptions unless a reference explicitly supplies them.
- Non-goals
 - Reimplement Internet routing.
 - Migrate a live TCP connection after proxy process loss.
 - Automatically retry arbitrary application writes.

Capacity and consequence — IC3 baseline

- Request/response bytes
 - At an assumed combined 2 KB/request, traffic is 200 MB/s.
 - Each payload crosses both a client-facing and backend-facing leg.
 - Account for both legs when sizing NIC and packet-processing capacity.
 - Decision: size on 400 MB/s of proxy I/O, not 200 MB/s, and treat packet rate as the likely first limit for small requests.
- Proxy concurrency
 - $100,000 \text{ requests/s} \times 0.1 \text{ s average duration} = 10,000 \text{ requests in flight}$.
 - At 32 KB of bounded buffering per request, buffers alone require about 320 MB fleet-wide.
 - Decision: bound request bodies and stream by default
 - A full-buffering default multiplies memory by body size.
- Health probes
 - $10 \text{ proxies} \times 500 \text{ backends} \div 2 \text{ s} = 2,500 \text{ probes/s}$.
 - Jitter probe schedules to avoid synchronized bursts.
 - Decision: keep per-proxy probing at this scale
 - Share coarse health only when $\text{proxies} \times \text{backends}$ makes probe traffic a visible backend cost.
- TLS
 - Handshake rate depends on connection reuse.
 - Benchmark both steady keep-alive and reconnect storms.
 - Decision: do not infer fleet size from request rate alone
 - A reconnect storm after a proxy loss is the CPU sizing case.

Architecture and flow — IC3 baseline

Data plane (per request):

```

Clients -> redundant front-door distribution -> L7 proxy fleet -> backend pools
                                     |
                                     local immutable route snapshot
                                     local health and in-flight state

```

Control plane (per configuration change):

```

Admin API -> config DB -> validator -> canary rollout -> proxy acknowledgments

```

- Each proxy owns live connection state.
 - Persistent configuration can be rebuilt.
 - Established TCP/TLS sessions cannot be reconstructed by another proxy just from configuration.
 - Cost: losing a proxy resets its connections
 - Clients must reconnect and in-flight writes become ambiguous.
 - A redundant front door distributes connections across proxy instances.
 - Managed load balancing is a valid deployment choice.
 - Discuss VIP failover or ECMP only if the interviewer asks how to build that layer.
 - Cost: the front door balances connections, not requests, so proxies can still see uneven request load.
 - The configuration control plane validates and versions routes and pools.
 - Last-known configuration preserves forwarding during a configuration-service outage.
 - Expired credentials or a required security revocation can still force rejection.
 - Cost: a 202 on a change means accepted, not applied
 - Convergence is observable, not instant.
 - Backends own their own admission limits.
 - The proxy bounds in-flight work per backend
 - The backend still needs its own queue limits.
 - Cost: two limits must be tuned together or the stricter one silently dominates.
- Accept the client connection and authenticate TLS as configured.
- Pin one validated configuration snapshot for the request.
- Resolve the route and eligible backend set.
- Select a backend and reserve local in-flight capacity.
- Proxy with bounded buffers and an end-to-end deadline.
- Release the reservation on success, failure, or cancellation.
- Record a classified outcome for passive health evaluation.

API and data model — IC3 baseline

Surface	Contract	Failure rule
PUT /v1/pools/{id}	Backend membership, weights, health policy	Validate positive weights and bounded timeouts; If-Match version, 412 on stale edit
PUT /v1/routes/{id}	Host/path match to a pool	Reject dangling pool references in the same snapshot
POST /v1/backends/{id}:drain	Stop new assignments; return 202 and an operation ID	Idempotency key; a repeat returns the same operation
GET /v1/operations/{id}	Deployment/drain progress	Expose partial rollout rather than claiming atomic fleet activation
GET /v1/pools/{id}/health	Per-proxy observations and observed time	Stale health is visibly labeled

State	Key	Stored fields
Route	(tenant_id, route_id)	Host matcher, path matcher, pool ID, version
Pool	(tenant_id, pool_id)	Policy, health thresholds, connect/request deadlines
Backend membership	(pool_id, backend_id)	Address, capacity weight, administrative state
Runtime health	(proxy_id, backend_id)	Consecutive outcomes, ejection deadline, in-flight requests

- Reference DDL
 - Keys and uniqueness match the state table above and the [companion schema](#).
 - Runtime health (proxy_id, backend_id) and the per-proxy immutable snapshot live in proxy memory; only desired configuration and operations are relational.

```

CREATE TABLE lb_pools (
  tenant_id text NOT NULL, pool_id text NOT NULL,
  policy text NOT NULL CHECK (policy IN ('P2C','ROUND_ROBIN','WEIGHTED_ROUND_ROBIN')),
  healthy_threshold integer NOT NULL CHECK (healthy_threshold > 0),      -- consecutive successes to readmit
  unhealthy_threshold integer NOT NULL CHECK (unhealthy_threshold > 0), -- consecutive failures to eject
  max_ejected_pct integer NOT NULL CHECK (max_ejected_pct BETWEEN 0 AND 100),
  connect_timeout_ms integer NOT NULL CHECK (connect_timeout_ms > 0),
  request_timeout_ms integer NOT NULL CHECK (request_timeout_ms > 0),
  version bigint NOT NULL CHECK (version > 0),                          -- If-Match validator
  updated_at timestamptz NOT NULL,
  PRIMARY KEY (tenant_id, pool_id)
);
CREATE TABLE lb_backends (
  tenant_id text NOT NULL, pool_id text NOT NULL, backend_id text NOT NULL,
  address text NOT NULL, port integer NOT NULL CHECK (port BETWEEN 1 AND 65535),
  weight integer NOT NULL CHECK (weight > 0),
  admin_state text NOT NULL CHECK (admin_state IN ('ACTIVE','DRAINING','DISABLED')), -- desired; a probe
  never rewrites it
  PRIMARY KEY (tenant_id, pool_id, backend_id),
  UNIQUE (tenant_id, pool_id, address, port),
  FOREIGN KEY (tenant_id, pool_id) REFERENCES lb_pools
);
CREATE TABLE lb_routes (
  tenant_id text NOT NULL, route_id text NOT NULL,
  host_match text NOT NULL, path_match text NOT NULL, pool_id text NOT NULL,
  credential_ref text, credential_version bigint, -- reference into the secret path; never
  key bytes
  version bigint NOT NULL CHECK (version > 0), updated_at timestamptz NOT NULL,
  PRIMARY KEY (tenant_id, route_id),
  UNIQUE (tenant_id, host_match, path_match), -- one owner per match
  FOREIGN KEY (tenant_id, pool_id) REFERENCES lb_pools -- rejects a dangling pool in the same
  snapshot
);
CREATE TABLE lb_operations (
  tenant_id text NOT NULL, operation_id text NOT NULL,
  kind text NOT NULL CHECK (kind IN ('DRAIN','DEPLOY')),
  target_id text NOT NULL, idempotency_key text NOT NULL,
  state text NOT NULL CHECK (state IN ('ACCEPTED','RUNNING','SUCCEEDED','FAILED')),
  acked_proxies integer NOT NULL DEFAULT 0 CHECK (acked_proxies >= 0), -- partial rollout, not atomic fleet
  activation
  total_proxies integer NOT NULL CHECK (total_proxies >= 0),
  created_at timestamptz NOT NULL, updated_at timestamptz NOT NULL,
  PRIMARY KEY (tenant_id, operation_id),
  UNIQUE (tenant_id, kind, target_id, idempotency_key), -- a repeat returns the same operation
  CHECK (acked_proxies <= total_proxies)
);
CREATE INDEX lb_operations_open ON lb_operations (tenant_id, updated_at)
  WHERE state IN ('ACCEPTED','RUNNING');

```

- Not captured by the DDL
 - Consecutive outcomes, ejection deadlines, and in-flight counts are per-proxy memory keyed by (proxy_id, backend_id); GET /pools/{id}/health reads a proxy-reported projection with its observed time, not a table.
 - The eligible-array snapshot a request pins is built from these rows per configuration version and swapped atomically inside the proxy.
 - TLS private keys are fetched by credential_ref from the secret path; the database stores the reference and version only.
- Deviation from the companion
 - The file's top-level routes and pools replace the companion's lb_id-scoped listeners; lb_routes is lb_listeners without the parent load-balancer resource.
 - lb_backends adds tenant_id to the state table's (pool_id, backend_id) key so the pool foreign key is tenant-scoped.

- `lb_operations` is added for `POST /backends/{id}:drain` and `GET /operations/{id}`; the companion delegates this to the generic resource model.
- Keep administrative state separate from observed health.
 - `DRAINING` must not be undone by a successful health probe.
 - A backend certificate failure must not be overridden by an availability policy.
- Keep TLS private keys in a controlled secret distribution path.
 - Route metadata contains a credential reference and version.
 - Roll credential updates with an overlap window and expiry alarms.

Selection and concurrency — IC3 baseline

- Default: power of two choices over eligible backends.
 - Compare normalized in-flight requests, not just TCP connections.
 - HTTP/2 multiplexing makes connection count a poor proxy for active work.
 - A maintained eligible-array snapshot makes sampling constant time.
 - Refiltering all backends on every request would cost $O(N)$.

```
PICK(snapshot):
  if snapshot.eligible_count == 0: return NO_BACKEND
  if snapshot.eligible_count == 1: choose its only backend
  otherwise sample two distinct eligible backends
  choose lower in_flight / capacity_weight
  reserve local in-flight capacity
  if reservation loses a drain/capacity race, retry selection within a bound
  release reservation in a finally/completion handler
```

- The score is local to a proxy.
 - A fleet-wide exact minimum would add coordination to every request.
 - Local randomized choice trades perfect balance for cheap independent decisions.
- Consistency boundary
 - The snapshot is immutable for the request
 - A swap publishes a new array atomically.
 - In-flight counters are per proxy.
 - Ten proxies see ten partial views of one backend.
 - A fleet-wide backend concurrency ceiling requires backend admission or divided permits
 - Local counters alone cannot enforce it.
- Concrete example
 - Backends `A` and `B` with weight `1`, `A` at `40` in flight and `B` at `10`.
 - Sampling both picks `B`
 - Sampling `A` twice is excluded by distinct sampling.
 - Exact failure: a counter leak on cancellation makes a healthy backend look busy forever.
- Session affinity
 - Use an authenticated stable session key only when required.
 - A hot session can overload one backend.
 - Affinity does not preserve a failed backend's in-memory state.

Health, retries, and draining — IC3 baseline

- Active probes
 - Example settings: interval `2 s`, timeout `0.5 s`, three failures to eject.
 - Detection is roughly `4.5–6.5 s` from failure under regular scheduling.
 - Scheduling stalls and network saturation extend that time.
 - Require two successes and slow-start traffic on recovery.

- Passive observations
 - Separate connection failures from application 5xx responses.
 - A shared downstream failure can affect every backend simultaneously.
 - Bound ejection percentage to avoid self-inflicted removal of the whole pool.
- Retry contract
 - At most one retry inside the caller's remaining deadline for the baseline.
 - Retry connect failures only when request execution is known not to have begun.
 - Retry application writes only when the backend enforces idempotency.
 - Cap retry traffic, for example at 10% of admitted original requests.
 - Streaming bodies and partially returned responses may not be replayable.
- Drain sequence
 - Publish DRAINING and stop new assignments.
 - Let in-flight requests finish within the drain deadline.
 - Stop admitting new HTTP/2 streams and close idle connections.
 - Terminate remaining connections at the deadline.
 - Confirm zero owned traffic before removing the backend.

Failure, operations, and security

Failure	Blast radius	Detection	Degraded behavior	Recovery
Proxy dies	Its live connections	Front-door health and connection resets	Clients reconnect; writes remain ambiguous	Remove failed proxy and restore fleet headroom
Backend stalls	Its assigned traffic	In-flight age and timeout rate	Circuit limits stop queue growth	Eject cautiously and ramp recovery
All probes fail	Entire pool	Correlated failures across proxies	Explicit panic policy or 503	Check probe configuration and shared dependencies
One proxy partitioned from backends	That proxy's share of traffic	Its health view diverges from peers	It ejects locally; peers keep serving	Front door removes it when its own probe fails
Configuration service unavailable	New changes only	Version lag	Forward from last-known valid configuration	Resume ordered version delivery
Bad configuration reaches canary	Canary cohort	Error and skew delta versus the fleet	Halt rollout; canary rolls back to prior snapshot	Fix and re-validate before expanding
Bad certificate rollout	A route cohort	Handshake errors	Retain valid overlapping credential if policy permits	Roll back credential reference
Retry storm during partial outage	Surviving backends	Retry ratio above budget	Budget exhausts; failures return to clients	Raise budget only after capacity returns
Slow client	A bounded connection set	Buffer occupancy and write timeout	Backpressure or close that client	Release its buffers and upstream reservations

- Panic behavior is a product decision.
 - Availability-sensitive traffic may retry a last-known eligible set when health evidence is unreliable.
 - Known compromised or administratively drained backends remain excluded.
 - Security-sensitive traffic may return 503 instead.
 - [Envoy panic thresholds](#) illustrate configurable availability behavior.
- Observe
 - Proxy-added latency independently of backend latency.
 - In-flight requests and buffer bytes.
 - Retry amplification.
 - Healthy fraction by proxy and pool.
 - Configuration acknowledgments.
- Retry discipline

- One retry, connect-phase or idempotent only, inside the remaining deadline.
- A per-proxy retry budget stops retries from becoming the second outage.
- Clients receive the original failure once the budget is spent, with a retry hint.
- Tenant isolation
 - Routes, pools, and certificates are tenant-owned
 - A tenant cannot reference another tenant's pool in a snapshot.
 - Bound per-tenant in-flight and connection counts so one tenant's slow clients cannot exhaust proxy buffers.
 - Health observations are per pool
 - One tenant's outage never ejects another tenant's backends.
- Rollout and rollback safety
 - Validate routes offline.
 - Canary a proxy cohort.
 - Compare error rates and pool skew before expanding.
 - Keep enough capacity to lose one planned failure domain.
 - Rollback is a snapshot swap to the prior version
 - Connections are not touched.

Alternatives and IC4 stretch

Option	What it is	Strength	Weakness	Choose when
L4 forwarding	Connection/packet distribution	No HTTP parsing requirement	Limited application routing	TCP services or TLS pass-through
L7 proxy	HTTP-aware forwarding	Routes and request policies	Parsing, TLS, and connection state	This HTTP API design
Round robin	Cyclic assignment	Simple and low cost	Ignores variable work	Requests and capacities are uniform
Power of two choices	Randomized local load comparison	Handles skew cheaply	Local load estimate can lag	Variable request duration
Consistent hash	Affinity-based placement	Cache locality	Hot keys and rebalancing effects	Session or cache affinity is required

- Base choice: L7 proxy with power of two choices.
 - The deciding variable is whether request duration varies enough to make round robin skew visible.
- IC4 stretch — safety before availability
 - Rank safety before availability for security failures.
 - Choose panic behavior per route class.
 - A health-check error cannot authorize bypassing backend authentication.
- IC4 stretch — 10× load
 - Measure the limiting resource.
 - Packet rate may dominate small requests.
 - Encryption may dominate new connections.
 - Memory may dominate slow streams.
 - Scale based on that resource rather than one universal requests/s threshold.
- IC4 stretch — cost and blast radius
 - Separate tenant traffic into cells when shared ejection or TLS load becomes a large blast radius.
 - Limit each cell's backend inventory.
 - Route new traffic away from an overloaded cell.
 - Existing connections still require a reconnect contract.
 - Cost: cells strand capacity
 - A cell sized for its peak idles at its trough.

- IC4 stretch — configuration rollout
 - Distribute immutable configuration generations and swap the active snapshot atomically per proxy.
 - Expose desired versus applied version so operators see convergence, not a fleet-wide claim.
- Weakest assumption
 - Local in-flight counts approximate backend load
 - A backend slowed by a shared dependency looks equally loaded from every proxy until latency-based outlier detection catches it.
 - Mitigation: add bounded passive ejection on latency percentiles and cap the ejected fraction.
- Over-engineering warning
 - Do not promise transparent connection migration or globally synchronized least-load selection.

Level signal — IC3 baseline vs IC4 stretch

Phase	IC3 must land	IC4 adds	Separating signal
Requirements	L7 choice, TLS termination, retry-safety of writes, drain deadline	Per-route panic policy and tenant-scoped configuration	Negotiates the retry contract instead of assuming it
Estimates	In-flight count, buffer memory, probe rate	Names packet rate or handshakes as the real limit for the given payload	A number changes the sizing resource
Architecture	Front door, proxy fleet, config plane with one trace	Front door balances connections not requests; 202 means accepted not applied	Every component has a mechanism and a price
Deep dive	Power of two on in-flight, hysteresis, one bounded retry, drain	Local-view limits, counter-leak failure, correlated-outage ejection cap	Reasons about stale and partial evidence
Operations	Failure table, canary, last-known configuration	Cells, retry budget as an outage control, safety-first panic	Volunteers control-plane outage behavior
Wrap	Recap and local-versus-global tradeoff	Names local load estimation as the weakest assumption	Identifies the flaw before the interviewer does

Interview pushback and validation

- Q1: Are the proxies stateless?
 - **IC3 baseline**
 - Their configuration is reconstructible, but connections and TLS state are local.
 - **IC4 stretch**
 - Decide whether reconnects meet the service contract before adding state replication.
- Q2: Why not least connections?
 - **IC3 baseline**
 - Requests multiplex over connections and have different durations.
 - **IC4 stretch**
 - Include observed work cost or latency only with stability controls against feedback oscillation.
- Q3: Every backend returns 500
 - Should we eject all of them?
 - **IC3 baseline**
 - Diagnose a shared dependency and apply bounded ejection.
 - **IC4 stretch**
 - Distinguish application failure from security failure in panic policy.
- Q4: The interviewer now requires raw TCP with end-to-end TLS.
 - **IC3 baseline**
 - Move to L4 connection routing and remove HTTP path inspection.
 - **IC4 stretch**
 - Relocate tenant-aware request limiting to the application boundary.

- Q5: How do you retry a timed-out POST?
 - **IC3 baseline**
 - Only through a server-enforced idempotency contract.
 - **IC4 stretch**
 - Account for streaming body replay and exhausted client deadlines.
- Q6: What happens to a long stream during deployment?
 - **IC3 baseline**
 - Drain with a documented deadline and reconnect behavior.
 - **IC4 stretch**
 - Use application checkpoints if resumability is required.
- Q7 (premise reversal): Why a proxy fleet at all?
 - Give clients the backend list and balance client-side.
 - **IC3 baseline**
 - Clients then own health, retries, and TLS to every backend, and a bad client library becomes a fleet-wide incident.
 - **IC4 stretch**
 - Client-side balancing fits internal, trusted, homogeneous callers with a lookaside control plane
 - Public and multi-tenant traffic still needs the proxy boundary.
- Q8: The configuration service is down for an hour.
 - What breaks?
 - **IC3 baseline**
 - Nothing on the request path
 - Proxies forward from the last-known snapshot and edits return 503.
 - **IC4 stretch**
 - Certificate expiry and revocation are the exceptions, so alarm on credential age and keep an emergency credential path.
- Validation scenarios
 - Zero and one eligible backend.
 - Drain racing with selection.
 - Cancellation releasing in-flight counters.
 - Correlated probe failure.
 - Malformed snapshot rollback.
 - Proxy loss with an in-flight non-idempotent write.
 - Configuration service outage with a certificate near expiry.

Recall card

- "I choose L7 for HTTP routing and keep connection ownership explicit."
- "Selection uses a local eligible snapshot and bounded in-flight work."
- "Health checks, retry budgets, and draining prevent amplification."
- "Configuration survives outages locally; failed connections still reconnect."
- Wrap-up script
 - "The design is an L7 proxy fleet behind a redundant front door, with per-request immutable configuration snapshots and local power-of-two selection."
 - "The key invariant is that a request sees one consistent snapshot, drained backends receive no new work, and a retry never assumes the backend did nothing."
 - "The main tradeoff is local load estimation, which keeps selection coordination-free at the cost of a lagging view under skew."
 - "My weakest assumption is that in-flight counts track real backend load; I would validate latency-based outlier detection under a shared-dependency slowdown first."

References

- [Ranked question and report provenance](#).
- [OCI role framing](#).
- [Detailed API and schema companion](#).
- Reference check: 2026-09-09.
 - Official documentation supports the mechanisms cited next to each link.
 - Capacity values and architecture choices remain proposed interview assumptions.
- [OCI load-balancer configuration](#): backend sets and health policy.
- [Envoy outlier detection](#): active workload observations and ejection.
- [Envoy panic threshold](#): configurable unhealthy-pool behavior.